

Practical Work #6

Synthesis of quadripoles

Objectives:

- Identify the relationship between transfer function and quadripole parameters
- Synthesize passive networks in graphical-analytical form
- Perform canonical and non-canonical syntheses.
- Obtain transmission zeros using partial removal.
- Implement active structures from a passive filter design
- Verify that the network obtained from the synthesis complies with the mathematical model and its physical behavior.

Deliverable exercises: only those exercises that appear with that label should be submitted.

Group: groups will be a maximum of 4 people

Approval condition: the practical work must be submitted up to 2 weeks after it is carried out in class.

Responsible: each practical work must have a person responsible. Submissions without the corresponding cover will not be accepted.

1) Synthesize a RC quadripole in open circuit that meets:

$$-I_{21} = \frac{s(s+1)}{s+2}$$

$$\text{and } \rightarrow \frac{s(s+2.25)}{s+2}$$

$$Y_{11} = \frac{(s+1)}{(s+2.25)} \text{ for } I_1 = 0$$

2) Synthesize a quadripole that meets:

$$Z_a = -\frac{s}{s+2}$$

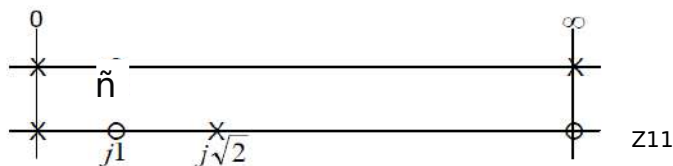
$$Z_{\infty} = \frac{1+2s^2}{s(s^2+2)}$$

3) Synthesize a quadripole that meets:

$$Y = -\frac{s}{s(s+4)}$$

$$\text{and } -\frac{(s^2+1)(s''+9)}{s(s''+4)}$$

4) Synthesize as a symmetric lattice



5) Knowing the transfer:

$$V_1 = \frac{s^2+1}{2s+1}$$

Y11

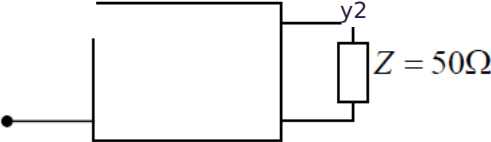
- Obtain a circuit with 3 inductors and 1 capacitor.
- Obtain a circuit with 3 capacitors and 1 inductor.

6) Synthesize a quadripole that meets:

$$Y_{11} = \frac{3s(s''+7\%) }{(s^2+2)(s^2+5)}$$

$$Y_{21} = \frac{s(s^2+1)}{(s^2+2)(s^2+5)}$$

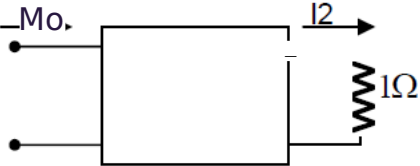
7) Synthesize the following non-dissipative transfer function:

$$\left| \frac{y}{y_1} \right| = \sqrt{\frac{1}{1 + w^6}}$$


8) Synthesize the following RC quadripole:

$$-\frac{I_2}{I_1} = H \cdot \frac{(5 + 9s + 14s^2)}{(s^2 + 8s + 12)}$$

$Z_1 = 6H$



9) Synthesize using a ladder circuit

$$T(s) = \frac{(s^2 + 4s + 9)}{(s^2 + 1)(s^2 + 3)}$$